

SPM-Spring 2026 - Oral Exam
Model (1)

1. Requirements elicitation may include:
 - A. Brainstorming and interviews
 - B. Coding and debugging
 - C. Deployment and testing
 - D. Database normalization
2. Which metric measures how frequently code changes occur?
 - A. Performance metric
 - B. LOC metric
 - C. Code churn
 - D. Deployment metric
3. Timing risk occurs when:
 - A. Early delivery always good
 - B. Timing has no effect
 - C. Delivery too early or too late makes system useless
 - D. Only budget is affected
4. Duration equation is —.
 - A. $D=E*M$
 - B. $D=E/M$
 - C. $D=M/E$
 - D. $D=M*E$
5. Using PERT method, assume an activity has the following three estimations: $a=6$, $m=12$, $b=18$. What is the standard deviation of the activity?
 - A. 2
 - B. 3
 - C. 4
 - D. 5

$b-a/6$

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Model (2)

1. Which artifact is interactive and used to test user experience?
A. Prototype B. Wireframe C. Mockup
D. CRUD matrix
2. Which architecture style separates services with independent lifecycles?
A. Pipe and Filter B. Monolithic Architecture
C. Microservices Architecture
D. Layered Architecture
3. Internal development (Make) means:
A. Developing in-house using organizational resources
B. Buying SaaS solution
C. Outsourcing to vendor only
D. Free downloading software
4. An activity requires Effort: 150 hours and is assigned to 3 resources, each available at 50% capacity. What is the duration of the activity in hours?
A. 40 B. 60 C. 80 D. 100
 $3 \times 0.5 = 1.5$
 $150 / 1.5 = 100$
5. In FP estimation, what is the maximum value of VAF?
A. 0.65 B. 0.135 C. 1.35 D. 135

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Model (3)

1. A disadvantage of plain-text requirements is:
 - A. They are too visual
 - B. They focus too much on user interaction
 - C. They may contain ambiguities
 - D. They cannot describe functions
2. A bug tracking system is used to:
 - A. Develop UML diagrams
 - B. Manage and track tickets
 - C. Replace testing
 - D. Deploy applications
3. React and re-plan refers to:
 - A. Adjusting plan when delays occur
 - B. Fixed planning
 - C. No monitoring
 - D. Initial estimation
4. The Function Point (FP) is calculated using —.
 - A. $FP = VAF / UFP$
 - B. $FP = VAF * UFP$
 - C. $FP = VAF + UFP$
 - D. $FP = VAF - UFP$
5. Using PERT method, assume an activity has the following three estimations: $a=6$, $m=12$, $b=18$. What is the variance of the activity?
 - A. 2
 - B. 3
 - C. 4
 - D. 5

$$(b-a/6)^2$$

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Model (4)**

1. Agile methodology is known for:
A. Linear development **B. Iterative development**
C. Hardware testing D. Waterfall processing
2. What is the main goal of deployment?
A. Writing requirements
B. Installing and making the system operational
C. Designing UML diagrams D. Monitoring bugs
3. Strategic alignment means:
A. Project matches organizational goals
B. Random feature addition C. Faster coding
D. Deployment method
4. An activity requires Effort: 180 hours and is assigned to 3 resources, each available at 100% capacity. What is the duration of the activity in hours?
A. 40 **B. 60** C. 80 D. 100
5. Using PERT method, assume an activity has the following three estimations: $a=6$, $m=12$, $b=18$. What is the duration of the activity?
A. 6 **B. 12** C. 18 D. 24

$$d = a + 4m + b/6$$

$$d = 6 + (4 \cdot 12) + 18/6$$

$$d = 12$$